

BOUNDARY-SPANNING FUNCTIONS SERVED BY RURAL SCHOOL PSYCHOLOGISTS

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Summary: The impact of perceived environmental demands on the roles and job satisfaction of the school psychologist in rural school systems was explored. Superintendents rated their perceptions of the community context and the psychologist's involvement in interagency linking functions. School psychologists rated their performance of boundary-spanning functions and their job perceptions. Results indicated that certain perceived environmental pressures are related to the performance of boundary-spanning activities. Those who performed boundary-spanning functions are more satisfied with their jobs and more influential in determining their roles.

Although authors concerned with characterizing the role and functions of school psychologists have touched on their increasing involvement in community-related services, the performance of professional activities outside of the educational system has never been systematically explored. Previous efforts to describe the functions performed by the school psychologist have focused primarily on direct clinical or consultation services (Meacham & Peckham, 1978; Ramage, 1979). Recognizing that such broad distinctions might obscure important differences in the practice of school psychology in urban versus rural areas, Hughes & Clark (1981) found that the rural practitioner's role was more diverse than his or her urban counterpart, less oriented toward assessment-related activities, and more oriented toward system-wide and community-related issues. The school psychologist's role vis-à-vis the community has been discussed by others who encourage coordination of efforts, communication of information regarding needs and services, and the initiation of community involvement (Brantley, Reilly, Beach, Cody, Fields, & Lee, 1974; Plas, 1981; Reilly, 1969; Sepez, 1972). Although closer relationships between local mental health programs and school psychological services have been advocated (Fine & Epstein, 1969), they may also be problematic, and specific interventions have been suggested to remedy problems of professional competition, overlapping functions, and differing role expectations (Steinberg & Chandler, 1976). Thus, there are potential costs as well as benefits to performing community-related functions.

Consideration of the factors associated with this expansion of the school psychologist's role is important for two reasons. Role definition, specifically the limited autonomy school psychologists enjoy and the lack of support they receive from teachers, administrators, and the public (Meyers, Brown, & Coulter, 1982), was one of the primary professional issues identified at the Olympia Conference. One of the general solutions discussed was enhanced visibility outside of the school system as well as more role flexibility within it. A second professional issue that has surfaced recently is a concern for high levels of stress, dissatisfaction, and turnover among school psychologists that have been viewed as direct results of too limited role definition, high

direct care caseloads, and lack of external support and service alternatives (Meyers, Brown, & Coulter, 1982; Wright & Thomas, 1982). Additional role flexibility and congruence between ideal and actual role, negotiated with school administrators, might prevent or alleviate some of these consequences, but other obstacles to school psychologists' abilities to achieve a broader role definition have been identified as well, such as their personal comfort with the usual assessment functions and their lack of skill or training in performing other functions. Thus, it is clear that we have only begun to explore the factors that impact the role of school psychologists.

The professional role of the psychologist, rather than being viewed as influenced primarily by the previous training of the psychologist or by service needs, is envisioned as part of a broader organizational structure that is influenced to varying degrees by outside pressures on school administrators. Thus, the defined roles of the school psychologist may be determined by the way administrators cope with these pressures and delegate certain functions. This examination will concentrate on the influence of external demands on school administrators and eventually on the functions performed by school psychologists.

The framework employed conceptualizes the structure of a school system and, therefore, the functions performed by school personnel as more responsive to environmental expectations and demands than to student needs or instructional processes (Meyer & Rowan, 1978). As external demands from regulatory bodies or the community impinge on the system, school administrators act to regulate and manage these relationships and maintain organizational effectiveness. These environmental-management activities have been described in research on other types of organizations as "boundary-spanning" roles (Aldrich & Herker, 1977; Liefer & Delbecq, 1978). They seem to predominate in environmental contexts that are undergoing change or that are closely intertwined, such as the relationship between a school system and its broader community. Aldrich & Herker (1977) proposed two primary types of boundary-spanning functions: processing information about environmental dynamics or trends to enable the organization to adapt more readily, and representing the organization to the community. Liefer & Delbecq (1978) have identified a third function, that of ensuring a flow of resources to and from the system. The overarching goal is continued adaptation and effectiveness of the organization. In previous observations and investigations of boundary-spanning activities, persons performing these functions were usually more satisfied with their positions, but experienced added role ambiguity and conflict. These roles typically afford individuals more flexibility in defining their functions and can become quite powerful, pivotal positions in large organizations.

Boundary-spanning activities in a school system might include efforts by school board members to involve their constituents in decisions or activities, speeches by the school superintendent to key community service groups, or liaison activities among human service agencies by the school psychologist or social worker. Although boundary-spanning is typically a managerial function, these roles may be delegated to other organizational members whose position or expertise lends itself to their performance. The proportion of an organization's roles allocated to, or time devoted to boundary-spanning activities varies directly with the heterogeneity of demands, the degree of stability or instability, the amount of resources, and the concentration of power in the environment (Aldrich, 1979). Factors determining the extent to which external linkages are pursued are especially important in rural areas where coordination among limited existing services is necessary, given high levels of need for human services,

new waves of in-migration, shrinking revenue bases, and a relative dearth of services available (Flax, Wagenfeld, Ivens, & Weiss, 1978).

The present study explored the extent to which superintendents in rural school systems perceive the environment as uncertain or turbulent, and the extent to which they involve other school personnel, namely psychologists, in boundary-spanning functions. The consequences of delegating such functions to school psychologists (i.e., the extent to which those performing linkage functions were satisfied with their jobs or experienced role conflict or ambiguity) also were examined. Where school superintendents rated the environment as complex, unstable, rapidly changing, and domineering, school psychologists were expected to report more boundary-spanning activities and, concomitantly, to report higher levels of job satisfaction, role ambiguity, and need for collaboration with other professionals to perform their jobs.

METHOD

Sample

Superintendents and school psychologists from 24 school districts in western Pennsylvania volunteered to complete the survey questionnaires and to be interviewed by telephone. These districts were small and rural, with an average enrollment of about 1,916 students (range = 406 to 3,100). In about half of the school districts, the school psychologists were employed by and operated out of the regional special education units in Pennsylvania. A total of 16 superintendents returned the surveys (67%), and 18 questionnaires were received from school psychologists (75%). One to 2 weeks later, each person was interviewed by phone.

Questionnaires

Two questionnaires were employed.¹ The superintendent's questionnaire contained 32 items compiled from previous empirical studies defining perceived environmental uncertainty, dynamism (Aldag & Storey, 1980), and complexity (Khandwalla, 1977). The first section asked the superintendent to assess the extent to which alternative actions, their probabilities of occurrence, and their outcomes could be specified; the second and third sections asked respondents to characterize the perceived external context of the school system as predictable, volatile, complex, manageable, and so on. In Sections 1 and 2, responses were made on a scale from complete disagreement (1) to complete agreement (5); Section 3 was scaled from 1 to 7 using descriptive anchors representing opposite extremes of a continuum.

The questionnaire completed by school psychologists was derived from empirical studies by Jemison (1980) defining boundary-spanning roles across organizations. Thirteen boundary-spanning roles were described, and the respondents indicated the extent to which each was a part of their job. Only those questions found to be most consistently associated with the overall boundary-spanning factor in Jemison's study were used. The wording on some questions was changed to reflect school activities rather than tasks in business systems. The reliability coefficient previously computed for the instrument (Cronbach's alpha) was .89. Respondents indicated on a 5-point Likert scale the

¹Copies of the questionnaires used are available from the author.

degree to which these activities were *never* a part of their work to *almost always* a part of their work. The second section contained 10 items on job satisfaction, role clarity, role overload, and the need to cooperate with others. The response scale was a Likert type from 1 = never to 5 = almost always.

Interviews

The interviews consisted of several open-ended questions. Superintendents were asked if linkage functions were valuable and important, for examples of boundary-spanning activities they or other administrators engaged in as differentiated from those the school psychologist performed, and for estimates of the amount of time devoted to these activities. School psychologists were asked to describe the linkage functions they performed, their importance to the school district, the amount of time devoted to them, and the factors that facilitated their involvement in interagency contacts.

Data Analysis Procedures

Means and standard deviations were computed for each item on the perceived environmental uncertainty questionnaire and on the boundary-spanning and job attitudes questionnaire. Since the study examined the relationship between perceived environmental uncertainty and boundary-spanning functions, correlation coefficients were calculated using the Pearson Product-Moment procedure. Similarly, correlation coefficients were calculated between boundary-spanning and job attitude items among school psychologists.

Levels of statistical significance for *t* tests and correlations were set at $< .05$, with those for *t* tests being two-tailed, since no differences between the groups of psychologists were expected a priori, while a one-tailed test was used for the correlation coefficients, since it was hypothesized that the correlations between perceived environmental uncertainty and boundary-spanning and between boundary-spanning and job attitudes would be positive and high.

RESULTS

Superintendent Questionnaire

Mean responses to all items on the Perceived Environmental Uncertainty (PEU) Questionnaire indicated that superintendents did not perceive their decision-making context as turbulent or difficult. Rather, the environment was seen as quite stable, relatively free of periodic fluctuations, and controllable (Table 1).

School Psychologist Questionnaire

On the boundary-spanning items, item means for all school psychologists responding indicated that much of their time was spent in meetings with students and parents (Table 2), whereas little time was devoted to formal presentations to external groups or to monitoring external services.

Responses from school psychologists employed by school districts and those in regional units were compared to ascertain whether there were consistent, significant dif-

Table 1
Environmental Perceptions: Responses of School Superintendents ($N=16$)

Item	<i>M</i>	<i>SD</i>
Perceived Environmental Uncertainty Scale ^a		
1. Typically, it is difficult to determine exactly what alternatives I have.	2.0	1.0
2. Before I make a decision, I generally have a solid fix on the probabilities of various potential subsequent events that may be relevant.	4.3	.5
3. Even if I know in advance what the outcomes of my choice would be, it would be hard for me to say how satisfied I would be with each of those outcomes.	2.7	.4
4. The major sorts of events that might influence the outcomes of my decisions are generally pretty easy to specify.	3.7	1.3
5. The number of alternatives I have available is often immense.	2.3	1.4
6. If I knew in advance what kinds of relevant events might subsequently occur, I could pretty well tell what the outcomes of my decisions would be.	3.6	1.2
7. I often find that the events that influenced the outcomes of my decisions were not those that I previously thought would be relevant.	3.1	1.3
8. I'm generally not fully aware of all the alternatives that may be available to me.	2.6	1.4
9. Each of my decisions has so many consequences that it's difficult to give very much thought to any one of them.	2.1	1.2
10. It's not possible to say with any degree of confidence what the probabilities of occurrences are of various events that could influence the outcomes of my decisions.	2.5	1.2
11. A tremendous number of events could affect the outcomes of my decisions.	3.6	1.5
12. I generally simply have to rely on my intuition rather than "hard data" concerning the probabilities of occurrence of various events that could affect the success of my decisions.	2.3	1.1
Environmental Dynamism Scale ^a		
13. Offering new programs or services is risky because there's a good chance they will be obsolete before they pay for themselves.	2.4	1.3
14. There are some other programs or services that could serve as pretty good substitutes for ours.	3.3	1.2
15. Our service can be pretty well predicted by projecting past service trends.	3.7	.9
16. We have little reason to worry about major innovations by other schools.	3.5	1.3
17. In my area, anyone able to predict what's going to happen next year is probably just a good guesser.	1.9	1.1
18. One nice thing about my community is that I quickly get accurate feedback on how well my system (departments, programs, etc.) has performed.	3.3	1.4
19. There's a good chance that the programs or service we'll be providing in a few years will have little resemblance to our current programs.	2.3	.9
20. Programs or services are relatively difficult to copy.	2.4	1.0

(continued)

Table 1 (Continued)

Item		<i>M</i>	<i>SD</i>
21. When I make a decision I usually have to wait quite a while to see if it was right.		2.7	1.3
22. We have what might be called a captive market in terms of alternatives for programs or services.		3.4	1.1
Environmental Complexity/Volatility Scale ^b			
23. Homogeneous	-Heterogeneous	3.8	1.2
24. Dynamic	-Stable	6.1	.9
25. Safe, Secure	-Risky, Insecure	3.0	1.2
26. Unpredictable	-Predictable	4.9	.8
27. Rapidly expanding	-Stagnant or even shrinking	4.8	1.4
28. Strong periodic fluctuations	-No periodic fluctuations	5.3	1.0
29. Return on investment good	-Difficult to get investment returns	3.8	1.4
30. Technically sophisticated	-Technically unsophisticated	4.5	1.7
31. Context exerts tremendous influence	-Local environment can be controlled or manipulated	5.3	1.0
32. Severely restricted context of legal, social or political constraints	-Quite unrestricted, constraint-free context	4.6	1.4

^aScale = 1 (completely disagree) to 5 (completely agree).

^bScale = 1 (completely disagree) to 7 (completely agree).

ferences between them across the items (Table 2). There were significant differences between the linkage roles played by school psychologists employed by school districts and those employed by regional units on 4 of the 13 boundary-spanning items. School district psychologists reported spending more time performing linkage or boundary-spanning functions, monitoring services outside the school district, and planning services to pursue within the school district. District psychologists were also frequently involved in acquiring information from professional or community groups and transmitting it to appropriate staff within the school district. These functions seemed to be performed more often on an informal basis, with the primary linkages being related to clinical case disposition.

Interviews

These differences were substantiated from information obtained during the interviews. On the average, school district psychologists reported spending about 10–15% of their time in activities linking school and community services, whereas psychologists employed by the regional unit reported that none of their time was devoted to these activities. Most of the regional unit psychologists lived outside of the communities they served and were assigned to several school districts on an annual, rotating basis. Linkages with community agencies, staff development, and follow-up case management were not a part of their formal job descriptions, nor were these functions supported by their supervisors.

Although these functions were not formalized or seen as a high priority by either superintendents or psychologists, the school district psychologists interviewed felt that

Table 2
Boundary-Spanning Item Summary for School Psychologists ($N=18$)

Linkage Functions ^a	All Psychologists <i>M</i>	School District Psychologists <i>M</i>	Regional Unit Psychologists <i>M</i>	<i>t</i> test
1. Provide information on a <i>formal</i> basis to groups outside your school system that is intended to create a favorable image of your system and its programs.	2.1	2.7	1.5	4.60*
2. Provide information on an <i>informal</i> basis to groups outside your school system that is intended to create a favorable image of your system and its programs.	2.9	3.6	2.6	2.03
3. Have primary responsibility for deciding on the external resources necessary for supplementing or enhancing the programs offered by your school system (e.g., funds, equipment, special therapy or training, testing).	2.6	3.4	2.4	1.39
4. Have primary responsibility for deciding when to acquire and acquiring the external resources to supplement or enhance programs offered by your school system (e.g., special, training therapies, testing, funds, or equipment).	2.5	3.4	2.1	1.81
5. Decide what portions of information acquired from sources outside your system to transmit to others within the system who will make use of it.	2.8	3.4	2.6	1.53
6. Decide when to transmit to others in your school system information acquired in your dealings with sources outside the system.	3.2	3.9	2.5	3.14*
7. Decide to whom information received from outside your school system should be conveyed.	3.4	4.0	2.6	2.48*

(continued)

Table 2 (Continued)

Linkage Functions ^a	All Psychologists <i>M</i>	School District Psychologists <i>M</i>	Regional Unit Psychologists <i>M</i>	<i>t</i> test
8. Acquire information <i>formally</i> from specific individuals or groups outside your school system that is needed by your department within the system (e.g., professional organizations, community agencies, consultants).	3.1	3.4	2.6	1.64
9. Acquire information <i>informally</i> from specific individuals or groups outside your school system that is needed by your department within the system.	3.4	3.4	3.0	.87
10. Meet with students and their parents to recommend specific services or programs <i>within</i> your school system.	4.5	4.3	4.9	1.89
11. Meet with students and their parents to recommend specific services or programs <i>outside</i> your school system.	3.8	4.1	3.9	.42
12. Decide on what kinds of students and services or programs your school system will pursue (conduct or develop).	2.8	3.7	2.5	2.56*
13. Need to monitor the other services being offered in your community or in other districts in order to provide effective services in your own school system.	2.3	2.7	2.0	1.22

^aScale = 1 (never) to 5 (almost always).

*Differences significant at $p \leq .05$, two-tailed test.

linkage activities were very important to performing competently and effectively and that they were seen as important by school district administrators. Direct diagnostic assessments and placement recommendations were defined as primary responsibilities for both types of school psychologists, with school districts expecting follow-up counseling, teacher consultation, and direct intervention from their employees.

Five types of boundary-spanning activities were most frequently described by psychologists and superintendents. Working with the local mental health center and with child welfare agencies, participating in joint staffing meetings on clients receiving multiple services, and conducting inservices on specific types of problems for staff of community agencies were encouraged by superintendents as linking functions. Giving talks to local community service groups and occasionally serving on the advisory board of a local human service agency were seen as legitimate informal activities (i.e., conducted infrequently and not during school hours). Most superintendents viewed direct service functions such as testing, counseling, and teacher consultation as paramount, but not to the total exclusion of these indirect activities. Some administrators saw linking functions as their own or the principal's responsibility, but having psychologists spend up to 10–15% of their time engaged in case-related boundary-spanning activities was accepted by administrators, and most superintendents did not discourage psychologists from engaging in these activities on their own time. Psychologists who performed many of these functions attributed their involvement to living in the community, knowing their contacts personally, enjoying the opportunity to build professional networks, and having the administrative support and personal style to engage in these activities. Many indicated that they had always been "networking types" and, therefore, had much experience doing it. Those who did not perform linking functions attributed their low level of involvement to a lack of administrative support and personal interest.

Correlations of Perceived Environmental Uncertainty (PEU) and Boundary-Spanning

The analyses correlating superintendents' PEU and psychologists' boundary-spanning activities were performed in two steps. First, all PEU items were aggregated into the three scales constructed in previous research: uncertainty, dynamism, and complexity/volatility. The boundary-spanning items were aggregated into five scales reflecting the empirical factors derived in previous research: representing organization, controlling resource flow between organization-environment, controlling information transmission, acquiring information, making client contacts and service plans. The PEU scale scores were then correlated with the scale scores for the boundary-spanning scales (Table 3). Only one of these correlations achieved a level of statistical significance warranting closer examination: representing the organization was moderately correlated with perceived environmental complexity/volatility ($r = .55$; $p < .10$).

To explore the correlation between representing the agency and perceived environmental complexity/volatility, correlation coefficients were calculated between individual items on the latter scale and the scale score on the former (Table 4). Representing the school district to external groups seemed to increase as the environment was perceived as heterogeneous, unstable, technically sophisticated, domineering, and relatively constraint-free ($r = .44, .60, .43, .40, .44$) by the superintendents.

Table 3
Correlations between Perceived Environmental Uncertainty Scales
and Boundary-Spanning Scales

PEU Scales	Boundary Spanning Scales				
	Represent Organization	Control Resource Flow	Control Information Flow	Acquire Information	Make Client Contacts
Uncertainty	-.09	.08	.20	-.17	-.04
Dynamism	-.25	-.03	-.25	.02	.09
Complexity/ Volatility	.55*	-.07	.28	.15	-.13

* $p < .10$, one-tailed test.

Boundary Spanning and Job Attitudes

Finally, school psychologists' perceptions of their jobs were assessed and correlated with the boundary-spanning scale scores. There were no significant differences in job satisfaction, role clarity, or influence regarding their roles between school district and regional unit psychologists (Table 5). Psychologists who engaged in more boundary-spanning activities reported having greater influence in decisions defining their role ($r = .62$) and higher job satisfaction. They were slightly more aware of the need to cooperate with others ($r = .31$) in performing their duties and saw their roles as being clear but somewhat conflicting ($r = .21, .25$); however, these correlations were not statistically significant. Those engaging in linkage activities did not feel overloaded in performing their function ($r = .03$), as might be expected from previous studies.

Table 4
Correlations between Representing the Organization
and Environmental Complexity/Volatility

Complexity/ Volatility Items	Representing Organization Item Score
Heterogeneous	.44*
Dynamic	.60**
Risky	-.06
Unpredictable	-.20
Rapidly Expanding	.15
Periodic Fluctuations	.34
Poor Investment Returns	.21
Technically Sophisticated	.43*
Domineering	.40*
Unrestricted	.44*

* $p \leq .05$, one-tailed test.

** $p \leq .01$, one-tailed test.

Table 5
Summary of Mean Responses on Job Attitude Items: School Psychologists ($N = 18$)

Item	All Psychologists	School District Psychologists	Regional Unit Psychologists	<i>t</i> test*
Job Satisfaction Scale				
1. In general, I like working in my present position.	4.6	4.7	4.5	.81
2. What happens to this organization is really important to me.	4.8	5.0	4.6	2.05
3. I am satisfied with the respect I receive from people with whom I work.	3.8	4.1	4.1	.05
4. This job doesn't offer much opportunity for developing my skills and abilities.	1.4	1.4	1.8	1.24
Influence				
5. In general, I have a lot of influence in most of the decisions about what I do.	4.2	4.3	4.0	.62
Role Clarity Scale				
6. Most of the time, I know exactly what to do in my job, i.e., what is expected of me.	4.7	4.4	4.8	.70
7. I find that it is hard to tell what good job performance means in this position.	2.3	2.1	2.4	-.39
Role Conflict				
8. I find, quite often, that I can't satisfy everyone at the same time and that this conflict frequently poses problems for me.	3.0	3.4	2.6	1.64
Role Overload				
9. I never seem to have enough time to get everything done.	3.8	3.9	3.8	.19
Cooperation Imperative				
10. To do my job well, I need to cooperate and work directly with other people.	4.9	5.0	4.9	1.00

Note: Scale = 1 (disagree completely) to 5 (agree completely).

*None of the differences between these means were significant at $p < .05$, two-tailed test.

DISCUSSION

Although perceived environmental uncertainty was expected to be an important aspect of administrative pressures in rural settings, especially during times of fiscal retrenchment, increasing rural population and regulatory demands (e.g., PL 94-142), superintendents in this study did not perceive these descriptors as characterizing the environments of their school systems. The environment generally was not seen as complex or unstable.

The primary hypothesis that aspects of the school psychologist's role would be linked to the administrator's perceptions of the environmental context was supported only partially. Superintendents and other administrators do engage, to varying degrees, in boundary-spanning activities. Many also see these as legitimate functions of the school psychologist. The boundary-spanning functions actually performed by school psychologists are closely related to the typical assessment and clinical services, centering primarily on management of clinical cases, and inservice or consultation to staff of community human service agencies.

When the community environment was perceived by superintendents as lacking external resources or as being unstable and fluctuating, school psychologists were more likely to meet with students and parents to plan specific service programs. Conversely, the more heterogeneous and abundant these resources, the less likely the psychologist would serve these functions, perhaps because parents have more alternatives on which to base their own decision to seek help. Psychologists tend to engage in more representing functions when the environment is perceived as diverse and exerting pressure on the schools. However, they do this informally, as a personal interest. Finally, they do not exert control over the acquisition and transmission of information within the school district. These functions are left to the administrators.

On a more personal basis, engaging in these linking functions also was associated with higher levels of job satisfaction and perceived influence within the school system and community. Thus, enlarging the role of school psychologists to include community-oriented services does not seem to produce substantial negative consequences when these changes are congruent with the administration's definition of their role and with their personal style preferences. Furthermore, the preference among school psychologists for greater role flexibility and enlargement has potentially motivating effects without producing serious role strain.

Given the obstacles cited previously to a broad role definition for school psychologists, e.g., the school system's narrow emphasis on academic goals, the psychologists' personal preferences or lack of training, and lack of funds, these data support the action plans formulated at the Olympia Conference by demonstrating that there is room for school psychologists to negotiate with administrators for a broader role and to gain more public visibility for their skills. However, they also point to a major "start up" problem that must be managed. Performing these functions informally outside of school hours might get them accepted and set the stage for more formal role redefinition, as well as for additional service opportunities. Until the positive impact of these informal inroads into role enlargement is felt by administrators, the emphasis probably will continue to be on assessment and direct service, both of which have potential for community linkage in terms of follow-up service plans and interagency cooperation on multiproblem families.

REFERENCES

- Aldag, R. J., & Storey, R. G. (1980, August). *Perceived environmental uncertainty: Measurement and correlates*. Paper presented at the Academy of Management Conference, San Diego, CA.
- Aldrich, H. (1979). Spanning organizational boundaries. In H. Aldrich, *Organizations and environments* (pp. 243-264). Englewood Cliffs, NJ: Prentice-Hall.
- Aldrich, H., & Herker, D. (1977). Boundary-spanning roles and organization structure. *Academy of Management Review*, 2, 217-230.
- Brantley, J. C., Reilly, D. H., Beach, N. L., Cody, W., Fields, R., & Lee, H. (1974). School

- psychology: The intersection of community, training institution, and the school system. *Psychology in the Schools*, 11, 28-32.
- Fine, M. J., & Epstein, S. J. (1969). The school psychologist's contribution to the community mental health center. *Journal of School Psychology*, 7, 70-75.
- Flax, J., Wagenfeld, M., Ivens, R., & Weiss, R. (1978). *Mental health and rural America: An overview and annotated bibliography*. Washington, DC: National Institute of Mental Health.
- Hughes, J. N., & Clark, R. D. (1981). Differences between urban and rural school psychology: Training implications. *Psychology in the Schools*, 18, 191-196.
- Jemison, D. B. (1980, August). *An empirical identification of interorganizational boundary-spanning roles*. Paper presented at the Academy of Management Annual Conference, San Diego, CA.
- Khandwalla, P. N. (1977). *The design of organizations* (Appendix A). New York: Harcourt Brace Jovanovich, Inc.
- Liefer, R. P., & Delbecq, A. (1978). Organizational-environmental interchange: A model of boundary-spanning activity. *Academy of Management Review*, 3, 40-50.
- Meacham, M. L., & Peckham, P. O. (1978). School psychologists at three-quarters century: Congruence between training, practice, preferred role, and competence. *Journal of School Psychology*, 16, 195-206.
- Meyer, J. W., & Rowan, B. (1978). The structure of educational organizations. In J. W. Meyer (Ed.), *Environments and organizations* (pp. 78-109). San Francisco: Jossey-Bass.
- Meyers, J., Brown, D. T., & Coulter, W. A. (1982). Analysis of action plans. *School Psychology Review*, 11, 161-185.
- Plas, J. (1981). The psychologist in the school community: A liaison role. *School Psychology Review*, 10, 72-81.
- Ramage, J. C. (1979). National survey of school psychologists: Update. *School Psychology Digest*, 8, 153-161.
- Reilly, D. H. (1969). Goals and roles of school psychology: A community-based model. *Journal of School Psychology*, 7, 35-37.
- Sepez, P. V. (1972). School psychology and the community: An intersection of sets. *Psychology in the Schools*, 9, 370-374.
- Steinberg, M. S., & Chandler, G. E. (1976). Developing coordination of services between a mental health center and public school system. *Journal of School Psychology*, 14, 355-361.
- Wright, D., & Thomas, J. (1982). Role strain among school psychologists in the midwest. *Journal of School Psychology*, 20, 96-102.

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